

HDOC DAY-37

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Batch-15

Question Statement

Write a menu-driven C program using separate functions to:

- Convert a lowercase string to uppercase without using built-in functions.
- Reverse a string.
- Check whether a string is a palindrome.
- Exit the program.

Aim

To write a menu-driven C program to perform various string operations using separate functions.

Algorithm

1. Start.
2. Display the menu:
 - Convert lowercase to uppercase
 - Reverse string
 - Check palindrome
 - Exit
3. Enter the user's choice.
4. Enter the string.
5. Call the function according to the selected choice.
6. Display the result.
7. Repeat the menu until **Exit** is selected.
8. Stop.

Choice Input		Expected Output
1	hello	Uppercase: HELLO
2	hello	Reverse: olleh

Choice	Input	Expected Output
3	madam	Palindrome
3	hello	Not Palindrome
4	—	Exit

>C PROGRAM:

```

1 #include <stdio.h>
2 #include <string.h>
3
4 void uppercase(char s[]) {
5     int i;
6     for(i = 0; s[i] != '\0'; i++)
7         if(s[i] >= 'a' && s[i] <= 'z')
8             s[i] = s[i] - 32;
9     printf("Uppercase: %s\n", s);
10 }
11 void reverse(char s[]) {
12     int i, n = strlen(s);
13     printf("Reverse: ");
14     for(i = n - 1; i >= 0; i--)
15         printf("%c", s[i]);
16     printf("\n");
17 }
18
19 void palindrome(char s[]) {
20     int i, n = strlen(s);
21     for(i = 0; i < n / 2; i++)
22         if(s[i] != s[n-1-i]) {
23             printf("Not Palindrome\n");
24             return;
25         }
26     printf("Palindrome\n");
27 }
28
29 int main() {
30     char s[100];
31     int choice;
32
33     do {
34         printf("\n1. Uppercase\n2. Reverse\n3. Palindrome\n4. Exit\n");
35         printf("Enter choice: ");
36         scanf("%d", &choice);
37
38         if(choice > 4 && choice <= 0) {
39             printf("Enter string: ");
40             scanf("%s", s);
41         }
42
43         switch(choice) {
44             case 1: uppercase(s); break;
45             case 2: reverse(s); break;
46             case 3: palindrome(s); break;
47             case 4: printf("Exit\n"); break;
48             default: printf("Invalid choice\n");
49         }
50     } while(choice != 4);
51     return 0;
52 }

```

>OUTPUT:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day37.c -o day37
[priyadarshna@kali:~]$ ./day37
1. Uppercase
2. Reverse
3. Palindrome
4. Exit
Enter choice: 1
Enter string: hello
Uppercase: HELLO
1. Uppercase
2. Reverse
3. Palindrome
4. Exit
Enter choice: 2
Enter string: computer
Reverse: retupmoc
1. Uppercase
2. Reverse
3. Palindrome
4. Exit
Enter choice: 3
Enter string: nadam
Palindrome
1. Uppercase
2. Reverse
3. Palindrome
4. Exit
Enter choice: 4
Exit
[priyadarshna@kali:~]$
```